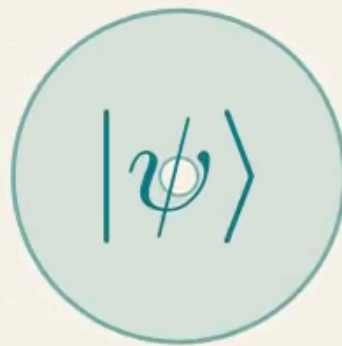




Day 5

Intro to Quantum Computing

KAUST ACADEMY + KGSP

Four days of how it works. Today: where it actually lives.

Four days of **how it works**. Today: **where it actually lives**.

No new algorithms. Instead: the machines, the cloud, the hype — and your next move.

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Those dates were not coordinated with your degree plan. Today ends with why they line up anyway.

Today, in four moves:

The machines — four ways to keep a qubit alive, and when you'd pick each.

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Where next — how to keep going after Friday.

And you're not touring empty-handed. This week you built:

✓ A CHSH game that beat the classical ceiling · a Bell pair · a teleporter.

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Every claim you hear today, you can check against something you personally ran.

The machines

Four ways to keep a qubit alive.

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Every machine you'll meet is one engineering answer to the same question: **how do you hide a qubit from the universe and still talk to it?**

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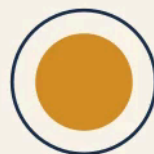
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(The trick works because of a threshold: once physical errors are rare **enough**, adding more qubits suppresses errors instead of adding them.)

now re-read the headlines: PHYSICAL counts, not logical ones



1 logical qubit

Shor on RSA-2048 wants $\sim 1,000$ logical $\rightarrow$ millions of physical qubits

today's machines: $\sim 100 - 1,000$ physical, $\sim$ a dozen logical (2026)

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The metrics that matter when you compare machines: how **long** qubits live, how **rarely** gates err, how **many** qubits, and **who can talk to whom**.

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When you'd pick it: you want many gates per second and the most mature cloud access — this afternoon's real job runs on one.

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When you'd pick it: deep circuits where every gate must count, and all-to-all connectivity saves you from SWAP overhead.

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When you’d pick it: big arrays for physics-like problems — quantum simulation of materials and spins.

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When you'd pick it: links between machines today; the bet is utility-scale fault tolerance later.

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No gates, no Grover, no Shor — QAOA's “gradual anneal” idea, built as dedicated hardware. Qubit counts from annealers and gate machines are **not comparable** — a favourite hype trick.

The milestone genre worth watching (you'll see these headlines for a decade):

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Every vendor roadmap now measures itself in logical qubits per year. When that number crosses 1,000, re-read Day 3's Shor slide.

The zoo, on one card:

family	superpower	weakness	pick it for
superconducting	speed	cold + neighbours-only	cloud access today
trapped ion	fidelity + all-to-all	slow gates	deep, precise circuits
neutral atom	array size	young gate set	simulating physics
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Nobody has declared a winner. Day 1's ENIAC story, replaying in real time — tubes vs relays, again.

The cloud

Nobody buys a dilution fridge. You rent minutes on someone else's.

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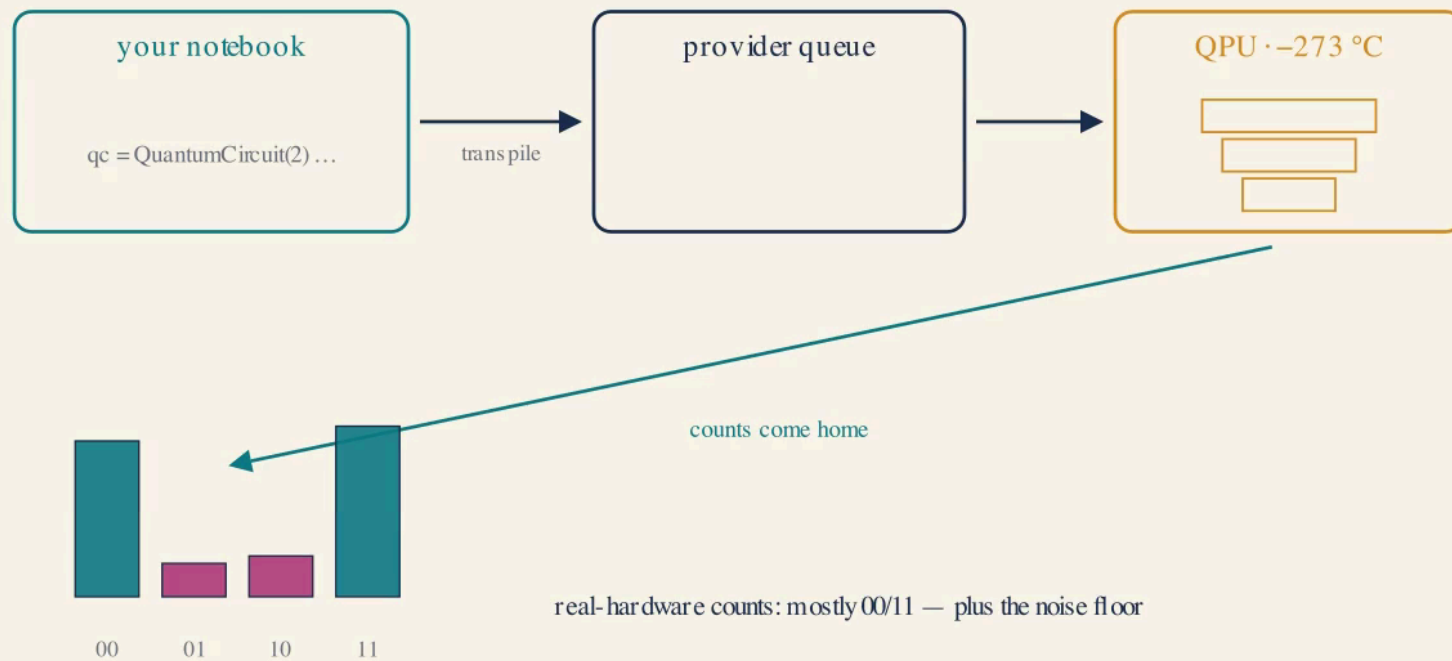
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All week your circuits ran on a local **simulator** — perfect, instant, free. Today one
of them graduates to the real thing.

the answer flies home as counts — noise fingerprints included



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(You’ve been calling `transpile(qc, backend)` all week. Now you know what it was doing — and why all-to-all connectivity was worth bragging about in the ion slide.)

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Queue: 46 jobs — the real cost of “free tier”.

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The habit: prototype on a simulator, book hardware minutes only when the circuit has earned them — machine time is billed by the task and queues are real.

So when do you actually **use** the real thing?

Simulator — while designing, debugging, sweeping knobs (everything this week); exact, instant, free... but tops out near 40 qubits, where 2^n eats your RAM.

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This afternoon you will personally cross it: a **56-qubit** circuit of yours runs where no laptop can follow.

This afternoon, a circuit of yours **queues on a real machine.**

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Prediction to hold: how will the real counts differ from the simulator's clean 50/50?
You've already met the culprit — it opened this morning.

In the field

Who uses this — and what they honestly get.

First, the honest market snapshot:

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Both facts are true at once. The industry is **pre-product and post-doubt** — paying now to be ready when the clocks strike.

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Track 1 of today's lab: VQE finds the hydrogen molecule's ground-state energy — real chemistry, two qubits.

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Track 2 of today’s lab: QAOA picks a portfolio — return up, risk down, brute-force honesty check included.

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Communications — QKD links and the future quantum internet (photonics' day job).

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Quantum software engineering — SDKs, compilers, cloud plumbing
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The field is young enough that its best tools don't exist yet — “the FORTRAN of quantum hasn't been written.” One of you might write it.

The hype filter

Reading quantum headlines without being fooled.

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“It tries all answers at once” — **no**: it measures **one**; interference has to be choreographed (you built that choreography — twice).

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“Quantum AI beats ChatGPT” — **no**: yesterday’s three costumes are the honest state of quantum ML.

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“Quantum supremacy/advantage” on a **useless benchmark** — winning a race only quantum computers wanted to run.

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“Which clock?” is the whole technique. Speaking of which...

The hype has bred its own backlash genre. This is a **real paper** (2025):

Replication of Quantum Factorisation Records with an 8-bit Home Computer, an Abacus, and a Dog

Peter Gutmann, University of Auckland
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Stephan Neuhaus, Zürcher Hochschule für
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March 2025, last update November 2025

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The lesson: sensationalism breeds equal-and-opposite skepticism. Neither the press release nor the dog is your guide — **the flags are. Stay in the calibrated middle.**

Green flags — the shape of **real** progress:

Logical-qubit milestones, with error rates **below** the raw hardware's — error correction actually winning.

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Boring engineering numbers trending: fidelities, coherence times, connectivity. The revolution will be incremental.

The hardest headline to judge: “*quantum computers will break encryption by 20XX.*”

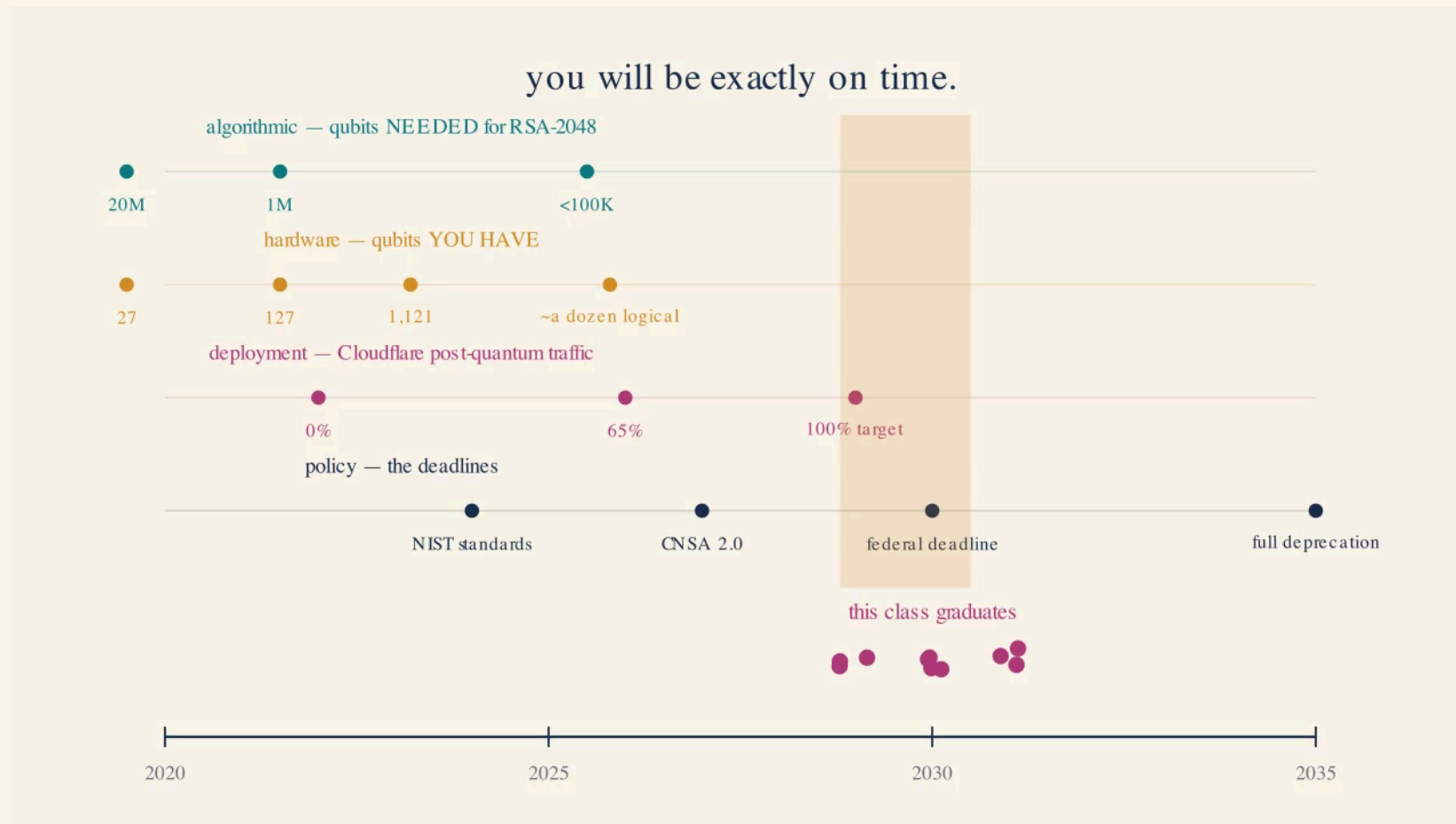
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Algorithm researchers, hardware roadmaps, internet operators, and governments. Watch what happens when you put them on one axis.



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deadlines land —

and your class walks into the job market in exactly that window.

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“Lab demonstrates logical qubit with $10\times$ lower error than its physical parts.”

(which flag list does this belong to — and why is it the bigger news?)

Where you go from here

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Want it to keep working? **Quantum Country** — a free quantum textbook with the same engine built in. Read it once; it schedules the reviews.

Three paths, all on your handout:

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Join — Qiskit's community Slack, Unitary Fund microgrants, arXiv quant-ph when you're ready to read the real thing.

If you want a concrete plan instead of a list — the next 12 months:

This summer: finish Quantum Country's first essay; let it quiz you (it will).

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Then: apply for QOSF mentorship with all of the above attached.

Everything from this block, on one page — **take it with you.**

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Plus the week's five ideas in five lines, for the day you want them back.

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Day 5 — real machines, real clouds, honest clocks — and your dates on the axis.

One lab left — and it ends past the simulator wall.

Track 1: VQE finds hydrogen's ground state.

Track 2: QAOA picks a portfolio.

Everyone: real IBM hardware — a GHZ hello, then the **π electrons of a 56-carbon polymer** — and Max-Cut on **the chip's own wiring**, tuned on toy sizes, deployed full-size.

Each track presents 3 minutes at the close.

Thank you for an excellent week. Go be on time.